INTRODUCCIÓ ALS COMPUTADORS

Laia Planas

1r Quatri

2026

07- 09 - 26

TEMA 2 - REPRESENTACIÓ DE NOMBRES NATURALS

Base 10 → 4322

$4 \cdot 1000 + 3 \cdot 100 + 2 \cdot 10 + 2 \cdot 1$

$4 \cdot 10^3 + 3 \cdot 10^2 + 2 \cdot 10^1 + 2 \cdot 10^0$

$X_u = \sum_{i=0}^{n-1} x_i 10^i$ (naturals)

Base 2 →

0 - 0
1 - 1
10 - 2
11 - 3
100 - 4

1100

$1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 0 \cdot 1$

$8 + 4 + 0 + 0 = 12$

$X_u = \sum_{i=0}^{n-1} x_i 2^i$

Digits = Bits

$X_u = \sum_{i=0}^{n-1} x_i b^i$

Base 16 → 0-9 A-F (10 - 16)

Hexadecimal

$0x\,23AF$

$2 \cdot 16^3 + 3 \cdot 16^2 + 10 \cdot 16^1 + 15 \cdot 16^0$

$X_u = \sum_{i=0}^{n-1} x_i 16^i$

3 dígits

[0 , 999]

[0 , $10^n - 1$]

$999 = 999 + 1 - 1 = \underbrace{1000}_{10^3} - 1$

rang $[0, b^n - 1]$

[0, 111]

[0, $2^n - 1$]

$111 = 111 + 1 - 1 = 1000 - 1 = 2^3 - 1$

Extensió de rang → 999 amb 5 dígits → 00999

Transformacions

b2 → b10

b2 → b16

b10 → b2

b10 → b16 → dividir a 2 i passar-ho a decimal

b16 → b2 → pàgina següent

b16 → b10

13 → binari

13 | 2
1 | 6 | 2
0, | 3 | 2
1, | 1

1101

8 4 0 1

F F

16 16

$15 \cdot 16^1 + 15 \cdot 16^0$

2 1 0

$7 \cdot 8^2 + 7 \cdot 8^1 + 7 \cdot 8^0$

4 3 2 1 0

$2 \cdot 3^4 \; 2 \cdot 3^3 \; 2 \cdot 3^2 \; 2 \cdot 3^1 \; 2 \cdot 3^0$

0000	0	
0001	1	
0010	2	
0011	3	
0100	4	
0101	5	
0110	6	
0111	7	
1000	8	
1001	9	
1010	10	A
1011	11	B
1100	12	C
1101	13	D
1110	14	E
1111	15	F

ABC1

1010 1011 0000 0001

0011 0101

0x 3 5

C37A

12 37 10

911

1214

E

56 14

Binari	Hexa	Dec
1101	D	13
0001 1111	1F	115
1110	E	14

477 |2
1, 238 |2
0, 119 |2
1, 59 |2
1, 29 |2
1, 14 |2
0, 7 |2
1, 3 |2
1, 1

64 |2
0, 32 |2
0, 16 |2
0, 8 |2
0, 4 |2
0, 2 |2
0, 1

1 1 1 1 1

$1\cdot2^4+1\cdot2^3+1\cdot2^2+1\cdot2^1+1\cdot2^0$

69 |2
1, 34 |2
0, 17 |2
1, 8 |2
0, 4 |2
0, 2 |2
0, 1

0100 0000

4 0

0100 0001

4 5

errors questionari ET2

Quin és el rang dels nombres naturals representables amb 2 digits codificats en Ox ?

- ~~0, 256~~
- 0, 255
- 0, 127
- 0, 128

FF

$15 \cdot 16 + 15 \cdot 16^0 = 255$

Representació en decimal de 1001 1011 ?

$1 \cdot 2^7 + 1 \cdot 2^5 + 1 \cdot 2^3 + 1 \cdot 2^1 + 1 = 155$

911 x

Representació decimal 1100 1110 ?

$2^7 + 2^6 + 2^3 + 2^2 + 2 = 206$

1214 x

Representació hexadecimal 0101 0110 1110

5 6 14 → E

5614 x

C37A → Binari

1100 0011 0111 1010 x

3CB → Binari

0011 1100 1011 x

Ox 4B → Decimal

411 x

$4 \cdot 16^1 + 11 \cdot 16^0 = 64 + 11 = 75$

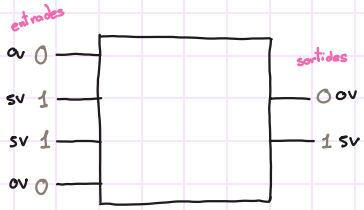
3DC → Decimal

31312 x

$3 \cdot 16^2 + 13 \cdot 16^1 + 12 = 988$

Tema 3 - Circuits Logics Combinacionals (CLC)

Donada qualsevol entrada, vull obtenir SEMPRE la mateixa sortida.

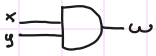


1 = 5V
0 = 0V

→ es defineix amb una taula de veritat

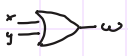
Components Bàsics

AND producte / multiplicar



x	y	w
0	0	0
0	1	0
1	0	0
1	1	1

OR suma ≥1



x	y	w
0	0	0
0	1	1
1	0	1
1	1	1

NOT nega el valor donat



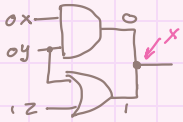
x	w
0	1
1	0

3 normes

1- No entrades lliures



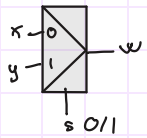
2 - No connectar sortides juntes



3 - No cicles



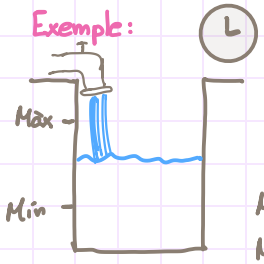
Multiplexor



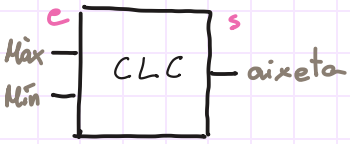
exercici 1

s	x	y	w
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Exemple:



Sensor Min { 0 Sec / 1 Moll

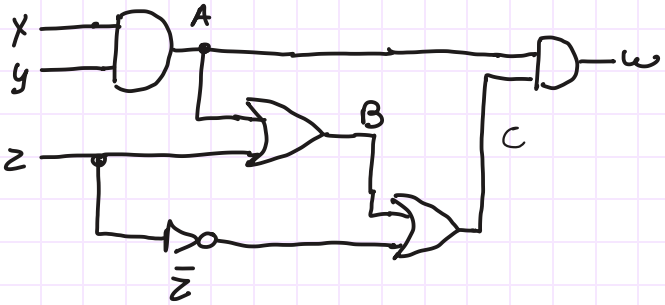


rellotge { 0 → nit / 1 → dia

aixeta { 0 → deixa passar aigua / 1 → tanca

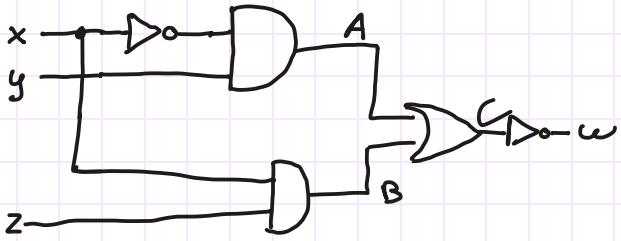
rell	mín	màx	aixeta	alarma
0	0	0	0	0
0	0	1	X	1
0	1	0	0	0
0	1	1	1	0
1	0	0	0	0
1	0	1	X	1
1	1	0	1	0
1	1	1	1	0

Exercici 2



x	y	z	z̄	A	B	C	w
0	0	0	1	0	0	1	0
0	0	1	0	0	1	1	0
0	1	0	1	0	0	1	0
0	1	1	0	0	1	1	0
1	0	0	1	0	0	1	0
1	0	1	0	0	1	1	0
1	1	0	1	1	1	1	1
1	1	1	0	1	1	1	1

Exercici 3



x	y	z	$\bar{x}$	A	B	C	w
0	0	0	1	0	0	0	1
0	0	1	1	0	0	0	1
0	1	0	1	1	0	1	0
0	1	1	1	1	0	1	0
1	0	0	0	0	0	0	1
1	0	1	0	0	1	1	0
1	1	0	0	0	0	0	1
1	1	1	0	0	1	1	0

3.

Questionari ET3a

- 1
- 1
- 1
- 1
- 1
- 0
- 1
- 1

- 0
- 0
- 1
- 0
- 1
- 0
- 1
- 0

a	b	c	$\bar{c}$	$\bar{b}$	D	E	f	g
0	0	0	1	1	0	0	0	1
0	0	1	0	1	0	0	0	1
0	1	0	1	0	0	0	0	0
0	1	1	0	0	0	1	1	0
1	0	0	1	1	0	0	0	1
1	0	1	0	1	0	0	0	1
1	1	0	1	0	1	0	1	1
1	1	1	0	0	0	1	1	0

x	y	z	A	B	v	w
0	0	0	1	1	0	0
0	0	1	1	0	0	0
0	1	0	0	0	0	0
0	1	1	0	0	0	0
1	0	0	0	1	0	0
1	0	1	0	0	0	0
1	1	0	1	0	0	0
1	1	1	1	0	0	0

Correcció

1- Un circuit CLC és aquell en el qual els valors dels senyals de sortida en un determinat moment depenent dels valors dels senyals d'entrada en aquest mateix moment. ✓

2-

x	y	z	f(x,y,z)
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

✓

3-

x_A	x_B	ig	s
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

✓

4- NAND w = 1,1,1,0 ✗
↳ NC NC

9- b → Si

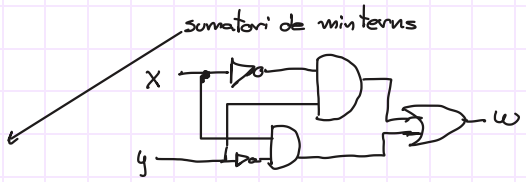
16-09-26

XOR

Minterms → la sortida dona 1

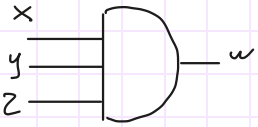
	x	y	w
M_0	0	0	0
M_1	0	1	1
M_2	1	0	1
M_3	1	1	0

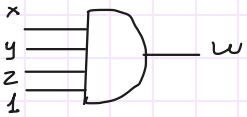
$M_1 + M_2$



Propietats AND i OR

- Commutativa
  - $x + y = y + x$
  - $x \cdot y = y \cdot x$
- Associativa
  - $x + y + z = (x + y) + z$
  - $x \cdot y \cdot z = x \cdot (y \cdot z)$
- Element neutre
  - $x + 0 = x$
  - $x \cdot 1 = x$

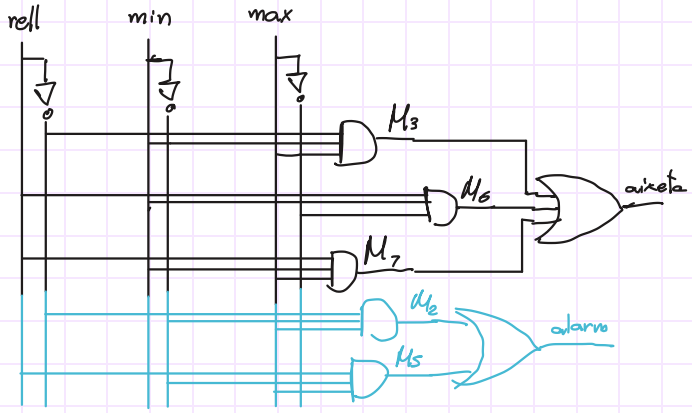




* $x=0$

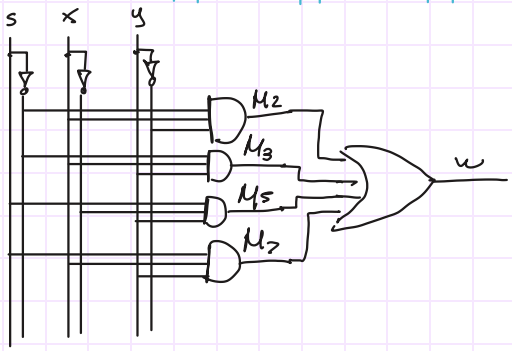
	rell	mín	màx	aixeta	alarm
	0	0	0	0	0
	0	0	1	X	1
	0	1	0	0	0
M_3	0	1	1	1	0
	1	0	0	0	0
	1	0	1	X	1
M_6	1	1	0	1	0
M_7	1	1	1	1	0

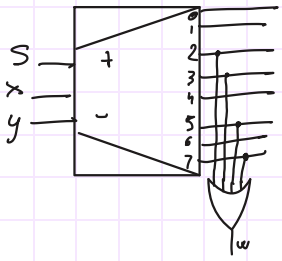
suma de minterms



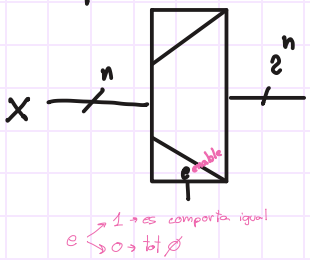
Exercici

	s	x	y	w
	0	0	0	0
	0	0	1	0
M_2	0	1	0	1
M_3	0	1	1	1
	1	0	0	0
M_5	1	0	1	1
	1	1	0	0
M_7	1	1	1	1

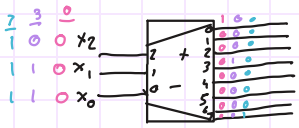


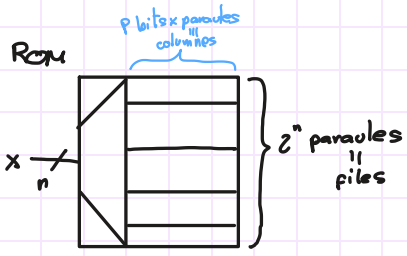


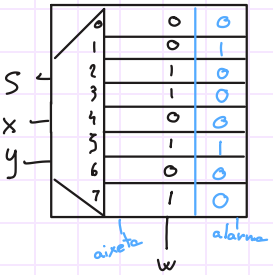
Decodificador



e → 1 → es comporta igual
e → 0 → bl Ø



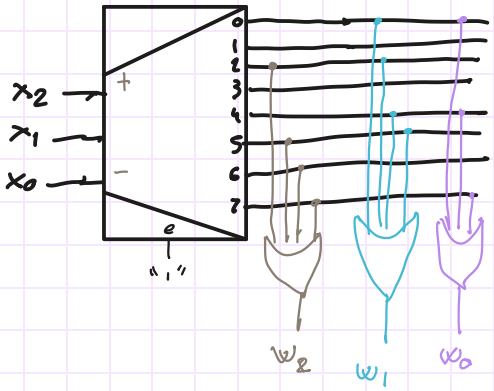




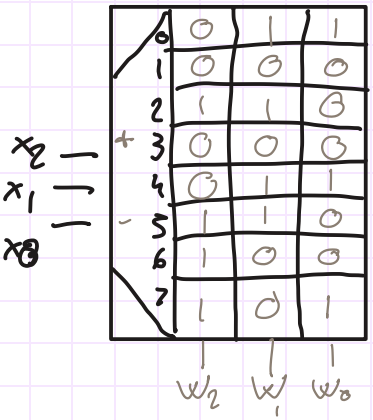
Exercici

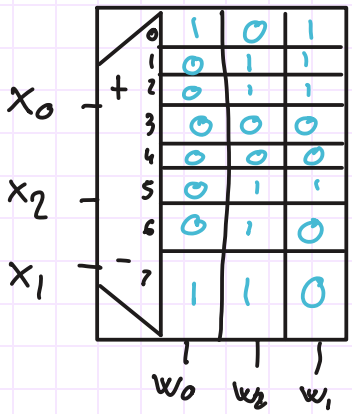
	x_2	x_1	x_0	w_2	w_1	w_0
M_0	0	0	0	0	1	1
	0	0	1	0	0	0
M_2	0	1	0	1	1	0
	0	1	1	0	0	0
M_4	1	0	0	0	1	1
M_5	1	0	1	1	1	0
M_6	1	1	0	1	0	0
M_7	1	1	1	1	0	1

Decodificador



ROM



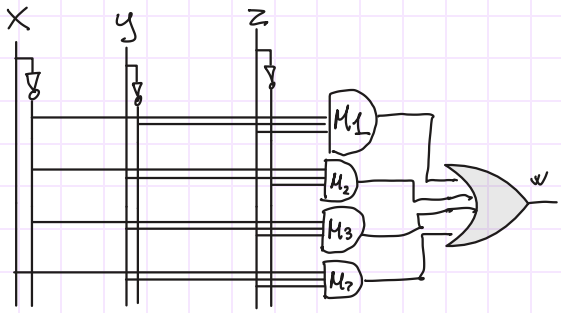


x_0	x_2	x_1	
0	0	0	0
1	0	0	4
0	0	1	1
1	0	1	5
0	1	0	2
1	1	0	6
0	1	1	3
1	1	1	7

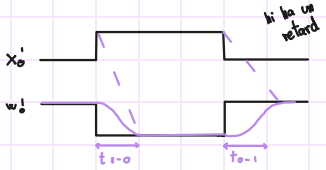
ET3b

2. $\bar{x}y\bar{z} + \bar{x}yz + x\bar{y}\bar{z} + xyz$

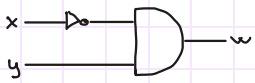
3. $\bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}yz + xyz$



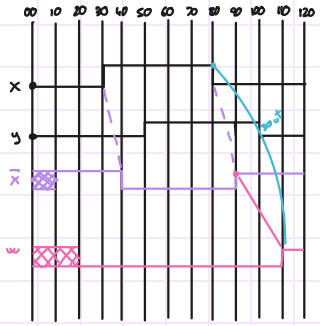
4. $\bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xy\bar{z} + xyz$



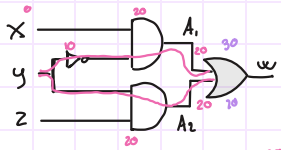
T_p (NOT) - 10 ut.
T_p (AND/OR) 20 ut.



T_p ( ) = 30 ut.



Exercicis

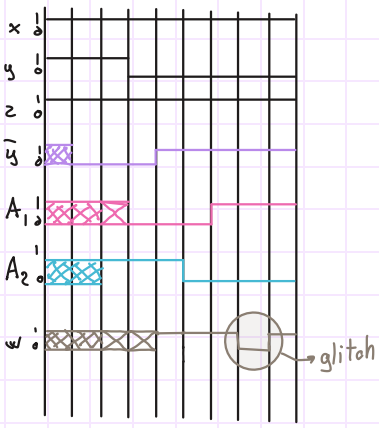


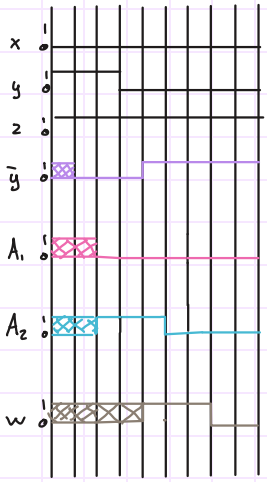
x – AND (20) – OR (20) – w – 40 ut

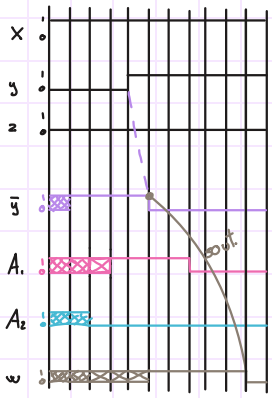
y – NOT (10) – AND (20) – OR (20) – w – 50 ut ← camí de propagació → T_p

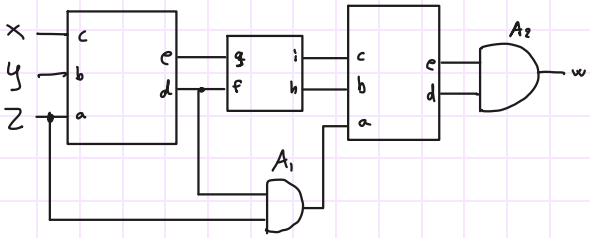
y – AND (20) – OR (20) – w – 40 ut

z – AND (20) – OR (20) – w – 40 ut









T_p	d	e
a	20	25
b	10	15
c	15	20

T_p	h	i
f	10	5
g	5	10

x – c (15) – d – f (10) – h – b (10) – d – A_2 (20) – w — 55 ut.

e – A_2 – w — 60 ut.

i – c (20) – e – A_2 (20) – w — 60 ut.

d – A_2 – w — 55 ut.

A_1 (20) – a (20) – d – A_2 (20) – w — 75 ut

e – A_2 – w — 80 ut.

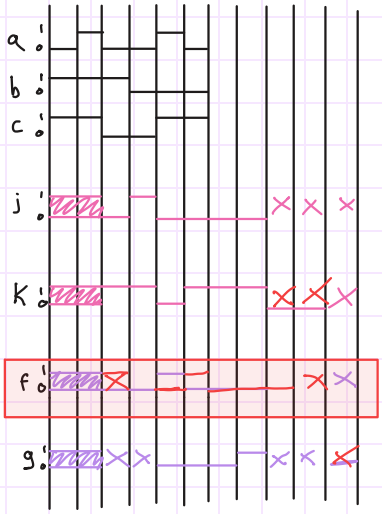
e (20) – g – i (10) – c (20) – e – A_2 (20) – w — 70 ut.

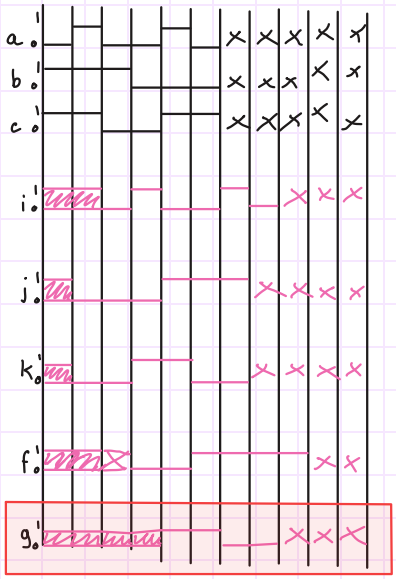
d – A_2 – w — 65 ut.

h (10) – b (15) – e – A_2 (20) – w — 65 ut.

d – A_2 – w — 60 ut.

ET3c





a- Mx -Dec- Not- f 80
a- Mx - Dec - g 70
b- XOR - Mx - Dec - Not - f 100
b- XOR - Mx - Dec - g 90
c- XOR - Mx - Dec - Not - f 100
c- XOR - Mx - Dec - g 90

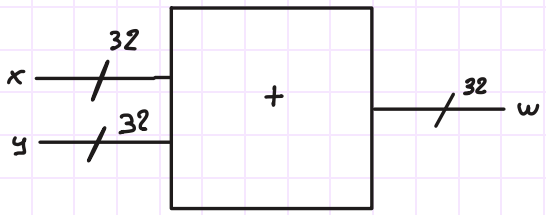
a- XOR - ROM - f 150 / XOR - OR - ROM 180
a- XOR - ROM - g 150
b- OR - ROM - f 150
b- OR - ROM - g 150
c- 140
~~150 / 130 / 160~~

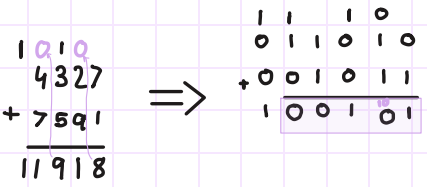
a - Mx - Dec - Not - f 100
a - Mx - Dec - g 90
b - Mx - Dec - Not - f 100
b - Mx - Dec - g 90
c- Dec - Not - f / Mx - Dec - Not - f
Dec - g / Mx - Dec - g 90
a - AND - AND - g 40
a - f N/A

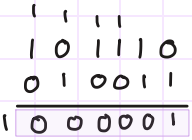
K - Dec - f 50
K - Dec - AND - g 70

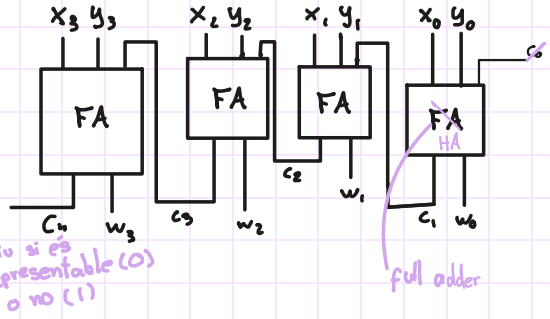
c - OR - Dec - f 70
c - OR - DEC - AND - g 90

23-09-26

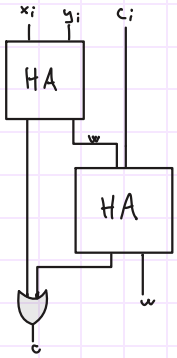


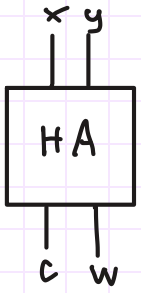






x_i	y_i	c_i	c_m	w_i
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



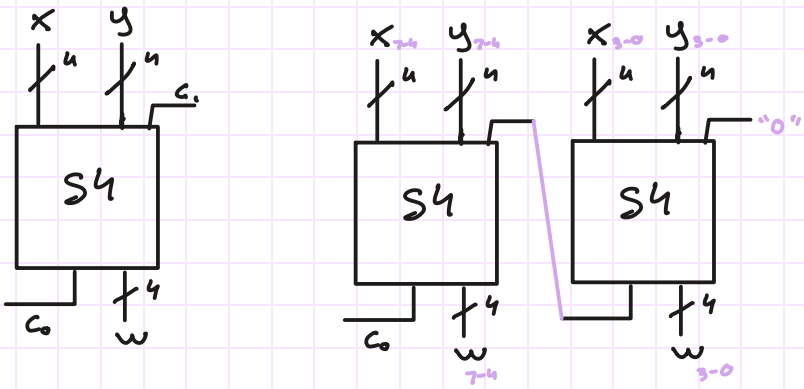


x	y	c	w
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

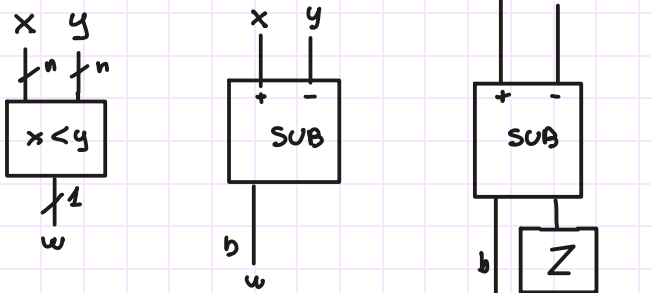


fer el restador → deures

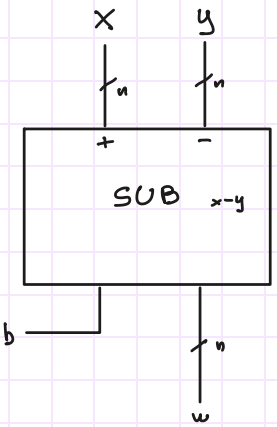
passar de 4 bits a 8 →

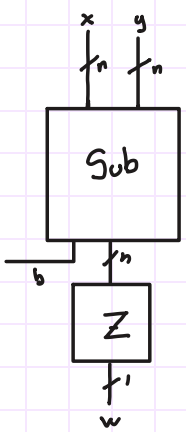


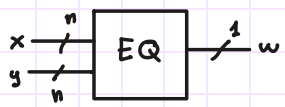
LTU → diu si $x<y$ LEU $x \leq y$



Comparador (EQ) que $x=y$?

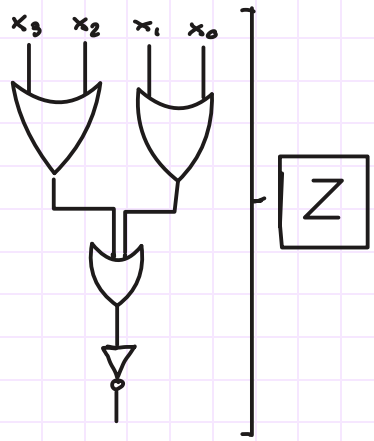






$$\begin{cases} 0 \text{ fals } (x \neq y) \\ 1 \text{ cert } (x = y) \end{cases}$$

$$Z \begin{cases} \text{si tots els bits d'entrades són } 0 \rightarrow 1 \\ \text{altrament} \rightarrow 0 \end{cases}$$



Components

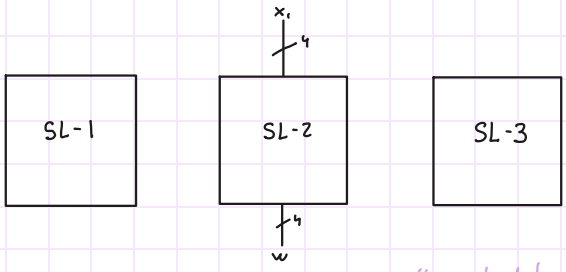
- ADD
- SUB
- EQ
- LTU
- LEU
- SR-X
- SL-X
- HA
- FA

$3251 \cdot 10^2 = 325100$

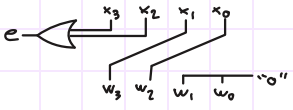
$1001 \cdot 4 = 1001 \cdot 2^2 = 100100$

$3251 / 10^2 = 325$ ✓

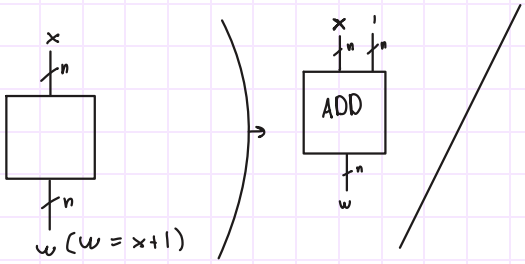
$10011 / 4 = 1001 / 2^2 = 100$ ✓

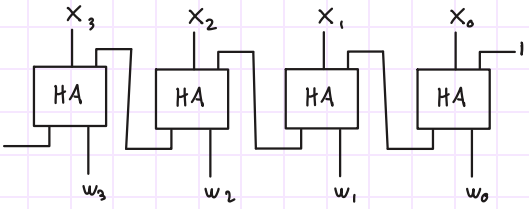


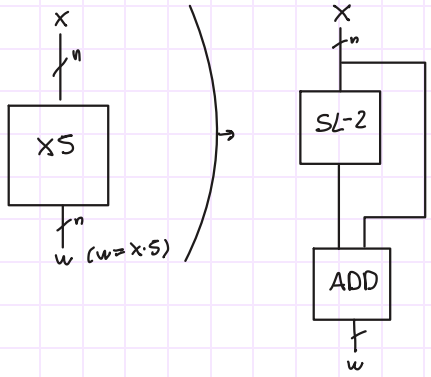
SR-2 → dos salts a la dreta



Exercicis







ET4



17 → LTU LEU

16 → 3

15 → 0

14 → 8

13 → 0

12 → 0100 R

11 → SI → No

8 → 0,1,1,0,1,0,0,1

7 → NR

Enters

28-09-26

Complement a 2

0	0	0	0
-7	1	0	0	1
-6	2	0	1	0
-5	3	0	1	1
-4	4	1	0	0
-3	5	1	0	1
-2	6	1	1	0
-1	7	1	1	1

$$X_u = \sum_{i=0}^{n-1} x_i 2^i$$

$$X_s = \begin{cases} x \text{ és positiu } (x_{n-1}=0) & \sum_{i=0}^{n-2} x_i 2^i \\ x \text{ és negatiu } (x_{n-1}=1) & -2^{n-1} + \sum_{i=0}^{n-2} x_i 2^i \end{cases}$$

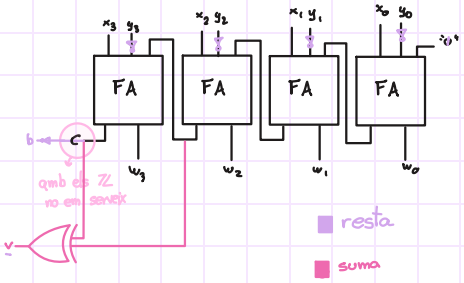
$$x_s = -2^{n-1} \cdot x_{n-1} + \sum_{i=0}^{n-2} x_i 2^i$$

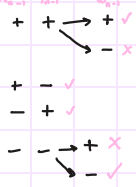
naturals $\mathbb{N} \Rightarrow [0, 2^n-1]$

enters $\mathbb{Z} \Rightarrow [-2^{n-1}, 2^{n-1}-1]$

0	0	0	0	000		
1	0	0	1	001		
2	0	1	0	010		
3	0	1	1	011		
-4	1	0	0	100		
-3	1	0	1	101		
-2	1	1	0	110		
-1	1	1	1	111		

$\mathbb{N}$ / $\mathbb{Z}$





x_{n-1}	y_{n-1}	c_{n-2}	c_n	w_{n-1}		v	
+	+	0	0	0	+	0	
+	+	0	0	1	-	1	
+	-	0	0	1	-	0	
+	-	1	1	0	+	0	
-	+	0	0	1	-	0	
-	+	1	1	0	+	0	
-	-	0	1	0	+	1	
-	-	1	1	1	-	0	

0 = ok
1 = no repres.

actua com un XOR

1 1 1 (-1)
0 0 0
1. 0↔1
0 0 1 (1) 2. +1

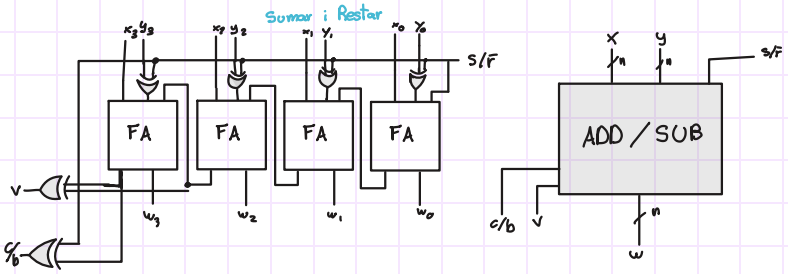
0 0 1 (1)
1 1 0
1 1 1 (-1)

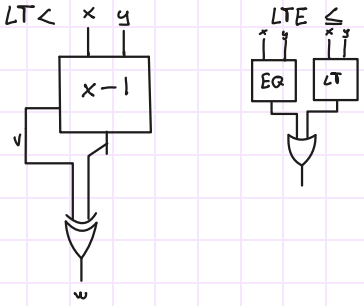
convertir el (5) en (-5). Nosaltres volem sumar, no restar

3 - 5 = 3 + (-5)

XOR

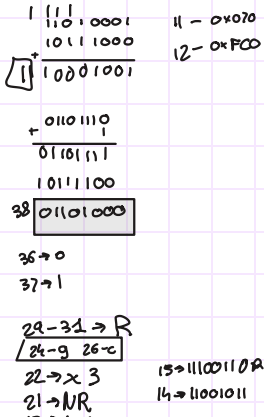
x	y	w
0	0	0
0	1	1
1	0	1
1	1	0



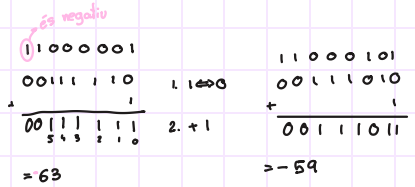


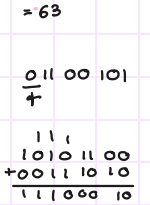


Buscar shift enters

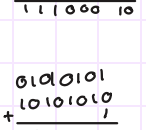


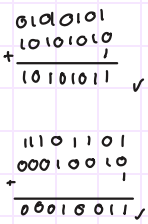
ET 5





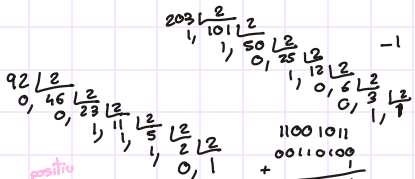
$X_U \rightarrow 2^6+2^5+2^2+1 = 101$ ✓

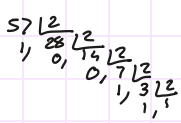


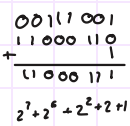




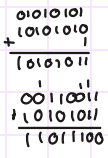


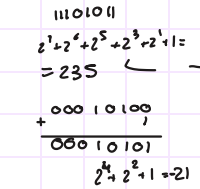




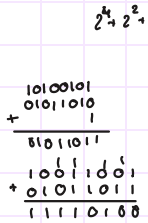


$2^7+2^6+2^2+2+1 = 199$

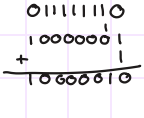


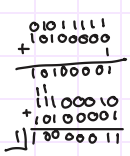


$2^7+2^6+2^5+2^3+2^1+1 = 235$

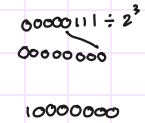


$2^4+2^2+1 = 21$

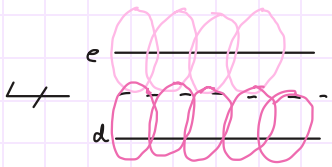


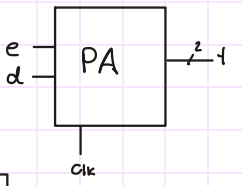


VR



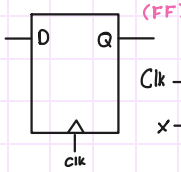
CIRCUITS SEQÜENCIALS



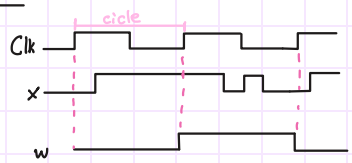


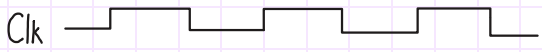
00
01 giro esquerra
10 giro dreta
11 —

Biestable D // Flip-Flop (FF)

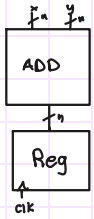


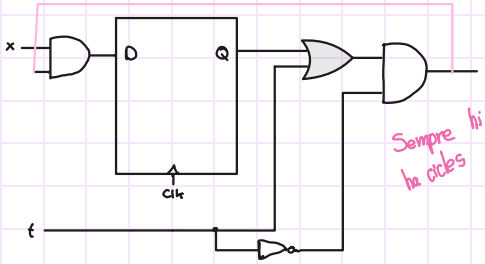
copiar de recordar les entrades d'un cicle i repetir-les





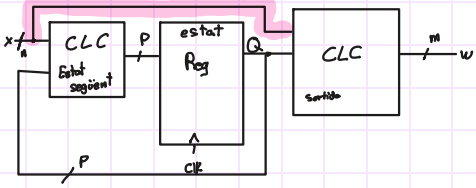
n biestables → Registre





Moore

sense la comanda espera 1 cicle més per donar la sortida



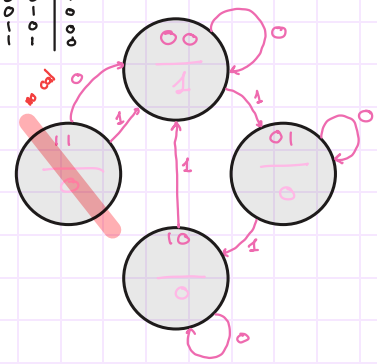
Tv estat següent
Tv sortida
Per definir el circuit

Tv estat següent

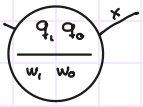
q_1	q_0	x	q_1^+	q_0^+
0	0	0	0	0
0	0	1	0	1
0	1	0	1	0
0	1	1	1	1
1	0	0	1	0
1	0	1	1	1
1	1	0	0	0
1	1	1	0	1

Tv sortida

q_1	q_0	w
0	0	1
0	1	0
1	0	1
1	1	0

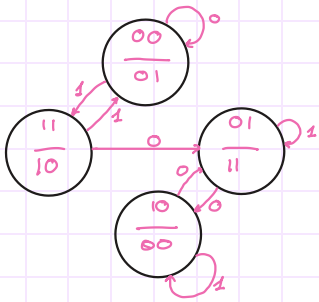


q_1	q_0	x	q_1^+	q_0^+
0	0	0	0	0
0	0	1	0	1
0	1	0	1	0
0	1	1	1	1
1	0	0	1	0
1	0	1	1	1
1	1	0	0	0
1	1	1	0	1

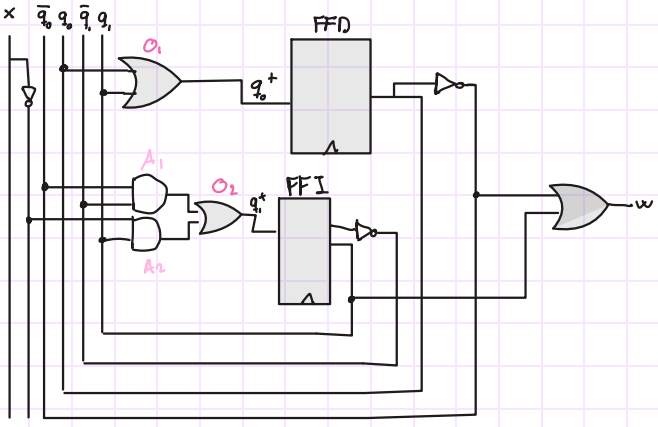


q_1	q_0	w_1	w_0
0	0	0	1
0	1	1	1
1	0	1	0
1	1	0	1

≡

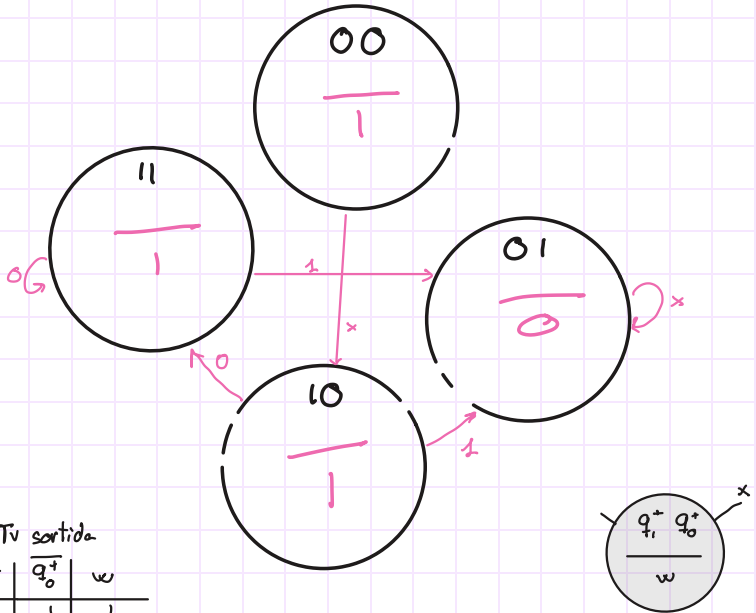


Exercici Proposat



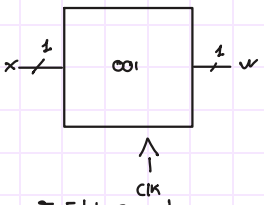
Tv entrada següent

q_{fa}	q_1	x	$\bar{q}_0$	$\bar{q}_1$	A_1	A_2	O_1	$\bar{O}_2$	q_1^+	q_0^+
0	0	0	1	1	1	1	0	1	1	0
0	0	1	1	1	0	0	0	0	0	1
0	1	0	1	0	0	0	1	0	0	1
0	1	1	1	0	0	0	1	0	1	1
1	0	0	0	1	0	0	1	0	1	0
1	0	1	0	1	0	0	1	0	1	1
1	1	0	0	0	0	0	1	0	0	0
1	1	1	0	0	0	0	1	0	0	1



Tv sortida

q_1^+	q_0^+	$\bar{q}_0^+$	w
0	0	1	1
0	1	0	0
1	0	1	1
1	1	0	1



x 0100001010010011101

w 0000000100001001000

T Estat Següent

q_1	q_0	x	q_1^+	q_0^+
0	0	0	0	1
0	0	1	0	0
0	1	0	0	0
0	1	1	1	0
1	0	0	0	1
1	0	1	0	0
1	1	0	0	0
1	1	1	0	0

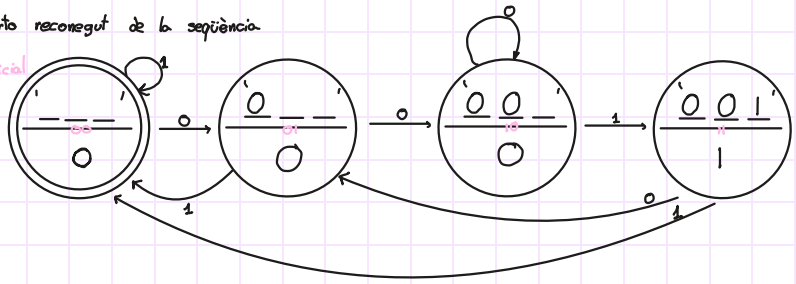
T sortida

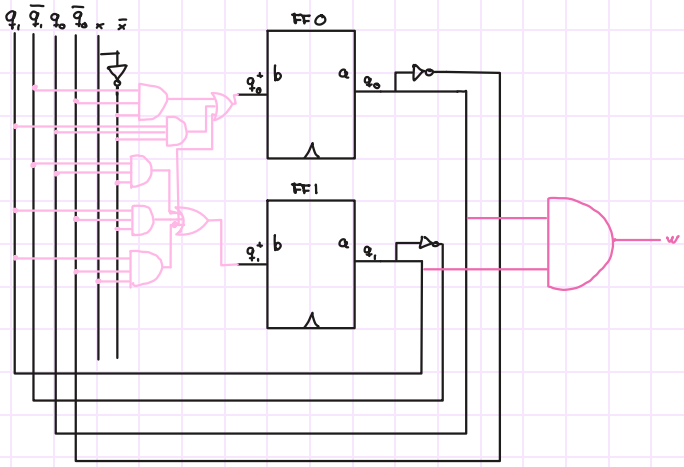
q_1	q_0	w
0	0	0
0	1	0
1	0	0
1	1	1

AND

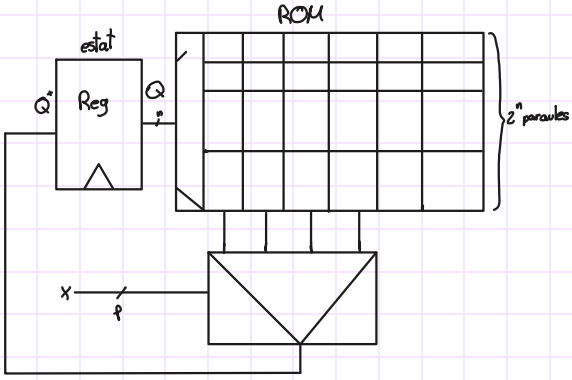


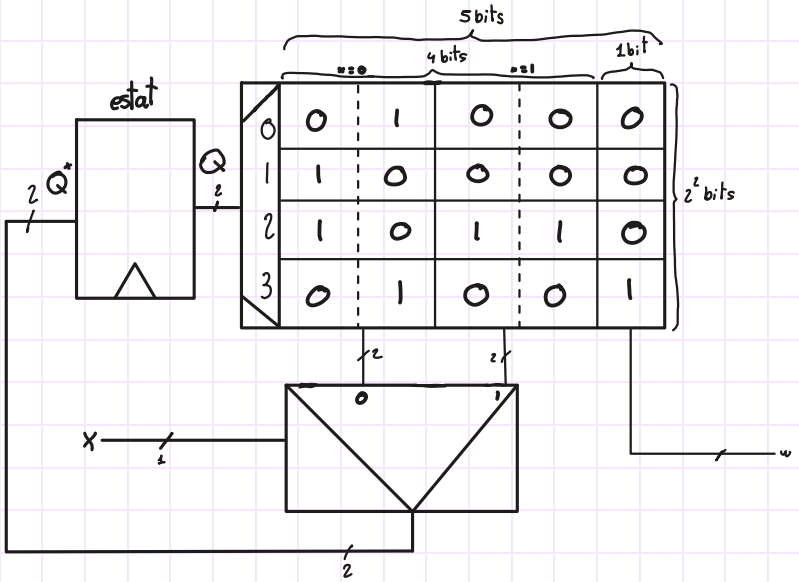
estat: què porto reconegut de la seqüència

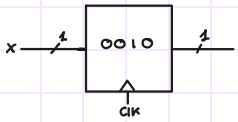




CIRCUIT ROM - Mux (multiplexor)

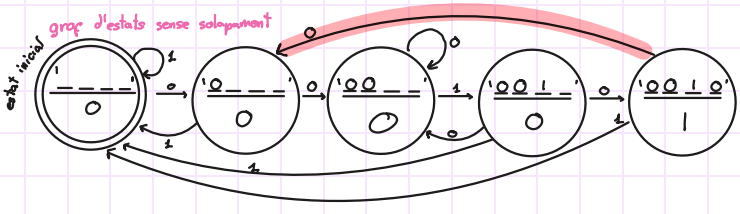




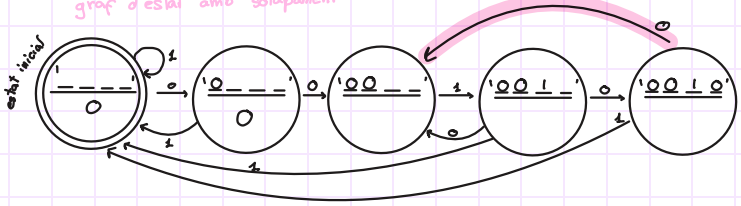


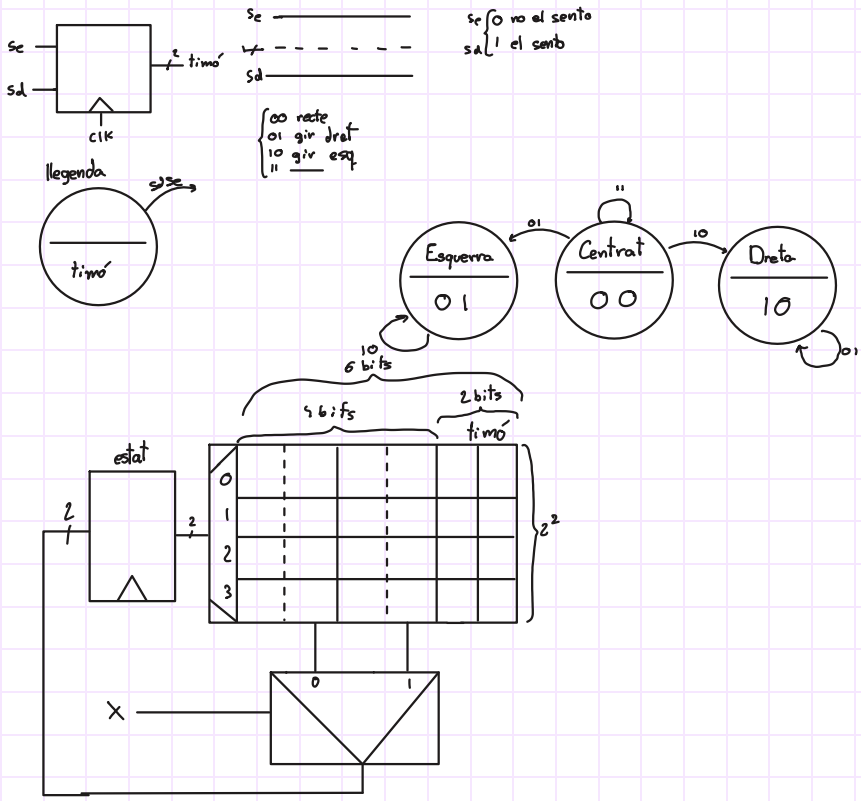
00010100100100010001001 sense solapament / amb solapament

graf d'estats sense solapament

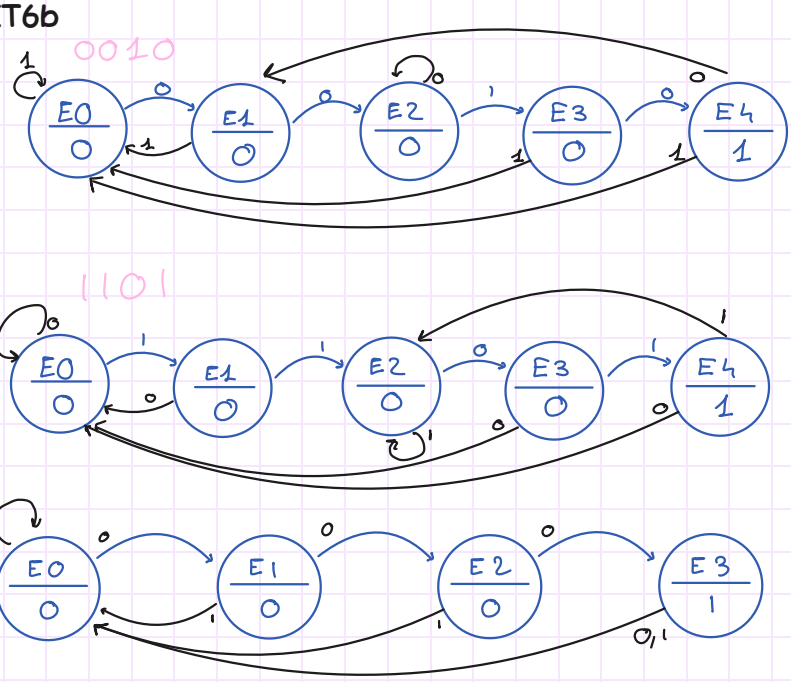


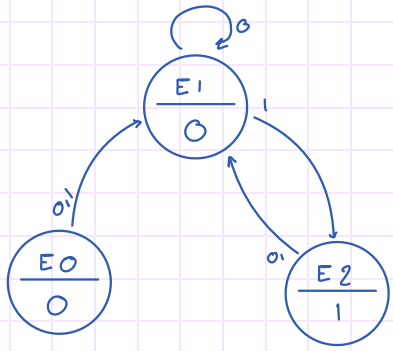
graf d'estat amb solapament





ET6b

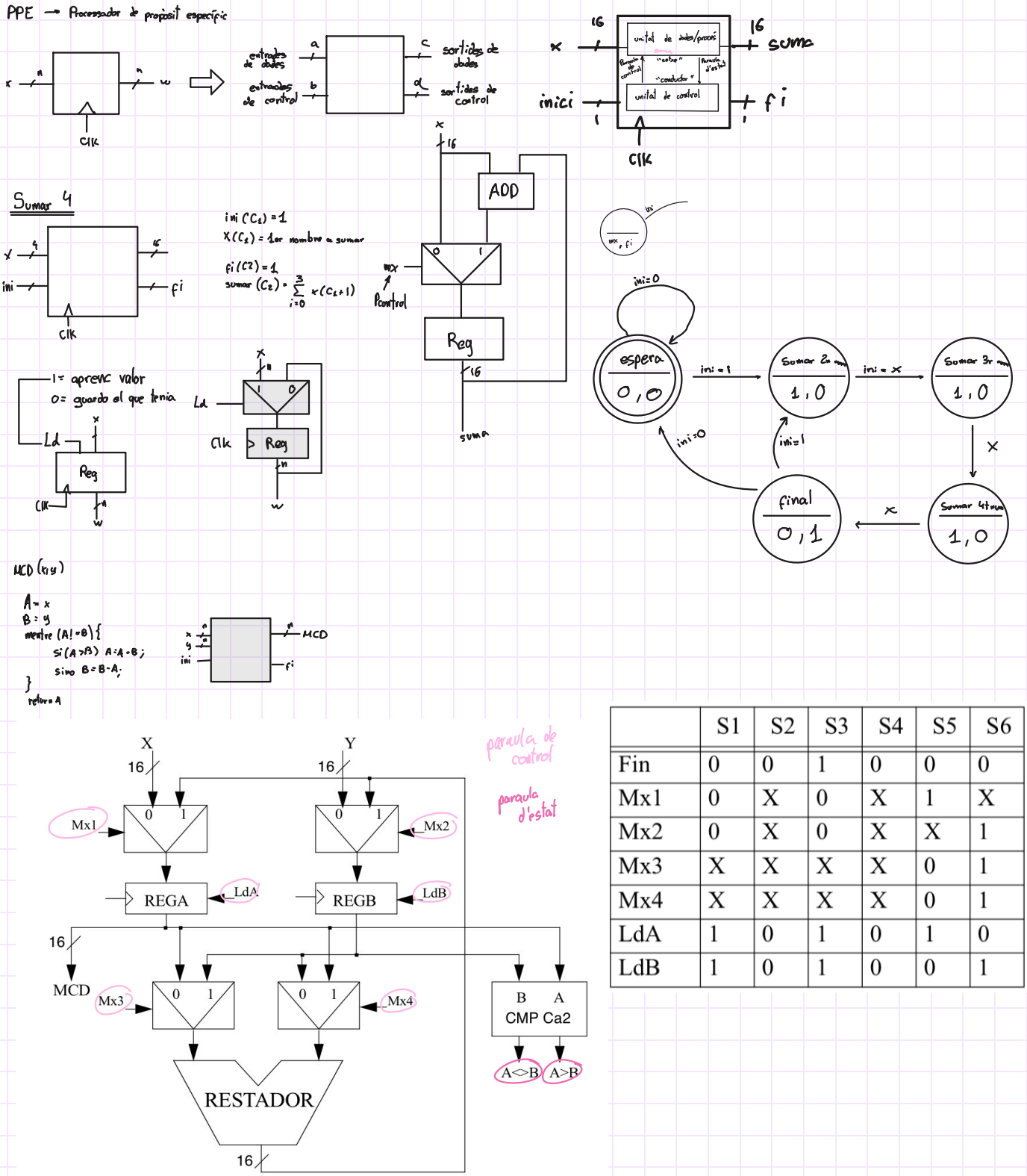




ET6c

in — C1 bd-FFO $\xrightarrow{Q0}$ C1ad

↳ C3 gj



	S1	S2	S3	S4	S5	S6
Fin	0	0	1	0	0	0
Mx1	0	X	0	X	1	X
Mx2	0	X	0	X	X	1
Mx3	X	X	X	X	0	1
Mx4	X	X	X	X	0	1
LdA	1	0	1	0	1	0
LdB	1	0	1	0	0	1

Figura 6. Unidad de proceso para el Máximo Comun Divisor.

